C PROGRAMMING

QUESTION BANK

1. Basic Programs

1. Write a C program to calculate Sum of two numbers. ***(Sum = a + b)***
2. Write a C program to find the area of a circle. ***(Area = π × r^2)***
3. Write a C program to calculate Volume of a Cube. ***(Volume = a^3)***
4. Write a C program to calculate Volume of a Cylinder. ***(Volume = π × r^2 × h)***

2. Conditional Statements

2.1 Simple If

1. Write a C program to check given number is positive, negative or zero.
2. Write a C program to check given number is even or odd.

2.2 If Else

1. Write a C program to find Largest of Two Numbers using if else.
2. Write a C program to find Largest of Three Numbers using if else.

2.3 Else if

1. Write a C program to check whether a number is positive, negative or zero.
2. Write a C program to read Roll Number, Name and Three Marks; Find Total, Percentage and Grade.

2.4 Nested If

1. Write a C program to find Largest of Three Numbers using Nested if.
2. Write a C program to find Smallest of Three Numbers using Nested if.

2.5 Switch Statement

1. Write a C program to check day using switch statement.
2. Write a C program to check month using switch statement.
3. Write a C program to perform arithmetic operations using switch statement.
4. Create a switch menu to check whether a number is even/odd or positive/negative/zero.

2.6 Conditional Operator

1. Write a C program to find Largest of Two Numbers using Conditional Operator.
2. Write a C program to find Largest of Three Numbers using Conditional Operator.

3. Write a C program to check whether a number is Odd or Even using Conditional Operator.

3. Jump Statement

3.1 goto Statement

1. Write a C program to print 1 to N using goto statement.
2. Write a C program to print N to 1 using goto statement.
3. Write a C program to print even numbers from 1 to N using goto statement.
4. Write a C program to find the sum of numbers from 1 to N using goto statement.

4. Looping Statement

4.1 While Loop

1. Print numbers from 1 to N and calculate their sum and average using while loop.
2. Write a C program to print N to 1 using while loop.
3. Write a C program to find factors, sum of factors and count using while loop.
4. Write a C program to check whether a number is prime or not using while loop.
5. Write a C program to check whether a number is perfect or not using while loop.
6. Print the digits of a number and find the sum of its digits using a loop.
7. Print the reverse of a number and check whether it is a palindrome using while loop.
8. Write a C program to check Armstrong number using while loop.
9. Write a C program to check Strong number using while loop.

4.2 do While

1. Create a menu using do-while and switch: check positive/negative/zero or even/odd.

4.3 For Loop

1. Write a C program to print 1 to N and find their sum using for loop.
2. Write a C program to print N to 1 using for loop.
3. Write a C program to check whether a number is prime or not using for loop.
4. Print the digits of a number and find the sum of its digits using a loop.

4.4 Nested for Loop

1. Write a C program to print prime numbers between two ranges using nested for loops.
2. Write a C program to print perfect numbers between two ranges using nested for loops.
3. Write a C program to print palindrome numbers between two ranges using nested for loops.
4. Print all three-digit Armstrong numbers between two given limits using nested for loops.

5. Write a C program to print Strong numbers between two ranges using nested for loops.

5. Array

5.1 1-Dimensional Array

1. Write a C program to find the sum of array elements using a 1D array.
2. Read array elements, print the even and odd numbers separately, and count each.
3. Write a C program to count positive, negative and zero elements using a 1D array.
4. Write a C program to search an element in a 1D array and count the number of times it occurs.
5. Read a 1D array and find the largest and smallest elements.
6. Read array elements and sort them in ascending and descending order.
7. Write a C program to separate even and odd numbers into two different arrays.
8. Write a C program to separate prime and non-prime numbers into two different arrays.
9. Write a C program to separate palindrome and non-palindrome numbers into two different arrays.
10. Write a C program to merge two arrays into a single array.

5.2 2-Dimensional Array

1. Write a C program to read a matrix, print it and find its transpose.
2. Write a C program to search an element in a matrix and count the number of times it occurs.
3. Read a square matrix and display its upper triangular and lower triangular matrices.
4. Read a square matrix, print its diagonal elements and find their sum.
5. Write a C program to add two matrices. Check whether rows and columns are equal.
6. Write a C program to check whether a matrix is symmetric or not.
7. Write a C program to multiply two matrices.
8. Read a square matrix and check whether all its diagonal elements are zero.

6. String

6.1 String Handling Functions

1. Read a string and demonstrate strlen(), strcpy(), strrev(), strcmp() and strcat().
2. Write a C program to check whether a string is palindrome using string functions.

6.2 Without Using String Functions

1. Write a C program to display a string and find its length without using string functions.
2. Write a C program to copy one string into another without using string functions.
3. Write a C program to reverse a string without using string functions.

4. Write a C program to check whether a string is palindrome or not without using string functions.
5. Write a C program to concatenate two strings without using string functions.
6. Read a string, print each vowel found, and count the occurrences of a, e, i, o and u.

7. Functions

7.1 User Defined Functions

7.2 User Defined Functions

7.2.1 Function No Argument & No Return Value

1. Write a C program to demonstrate a function having no arguments and no return value.
2. Write a C program to calculate the area of a circle using a function having no arguments and no return value.
3. Write a C program to calculate the factorial of a number using a function having no arguments and no return value.
4. Write a C program to find the largest of three numbers using a function having no arguments and no return value.

7.3 User Defined Functions

7.3.1 Function Argument & No Return Value

1. Write a C program to demonstrate a function having arguments and no return value.
2. Write a C program to calculate the area of a circle using a function having arguments and no return value.
3. Write a C program to calculate the factorial of a number using a function having arguments and no return value.
4. Write a C program to find the largest of three numbers using a function having arguments and no return value.

7.4 User Defined Functions

7.4.1 Function Argument & Return Value

1. Write a C program to demonstrate a function having arguments and a return value.
2. Write a C program to calculate the area of a circle using a function having arguments and return value.
3. Write a C program to calculate the factorial of a number using a function having arguments and return value.
4. Write a C program to find the largest of three numbers using a function having arguments and return value.
5. Write a C program to check whether a given number is prime or not using a function having arguments and return value. Return 1 if prime, otherwise return 0.

6. Write a C program to check whether a given number is palindrome or not using a function having arguments and return value. Return 1 if palindrome, otherwise return 0.

7.5 Array Function

1. Pass a 1D array to a function and calculate the sum of its elements.
2. Pass a 2D array to a function and display the matrix elements.

8. Structure

1. Read item code, name, quantity and price; calculate total price and display the item details.
2. Read one student's details, calculate total marks, percentage and grade, and display the mark list.
3. Read N students' details, calculate total, percentage and grade, search by roll number, and sort students by total marks.
4. Read one employee's details and calculate TA, DA, HRA, LIC, PF and net salary.
5. Read N employees' details, calculate allowances, deductions and net salary, search by employee ID, and sort by net salary.

9. Pointers

9.1 Pointer Programs

1. Write a C program to find the sum of two numbers using pointers.
2. Write a C program to find the largest of three numbers using pointers.

9.2 Passing to Function

1. Write a C program to find the sum of two numbers using pointers and function.
2. Write a C program to find the largest of three numbers using pointers and function.

9.3 Pointer to Pointer

1. Write a C program to find the sum of two numbers using double pointers.

9.4 Dynamic Memory Allocation

9.5 Dynamic Memory Allocation

9.5.1 1D Array Pointer

1. Write a C program to read array elements using dynamic memory allocation, display the array elements and find their sum.
2. Write a C program to read array elements using dynamic memory allocation and find the largest and smallest elements.
3. Read array elements using dynamic memory allocation, sort them in ascending order, and find the largest and smallest elements.

9.6 Dynamic Memory Allocation

9.6.1 2D Array Pointer

1. Read a matrix using dynamic memory allocation and display its transpose.
2. Read a matrix using dynamic memory allocation, print its diagonal elements and find their sum.
3. Write a C program to add two matrices using dynamic memory allocation.
4. Write a C program to check whether a matrix is symmetric using dynamic memory allocation.

9.7 Dynamic Memory Allocation

9.7.1 Structure Pointer

1. Read one student's details using a structure pointer, calculate total, percentage and grade, and display the mark list.
2. Read N students' details, calculate total, percentage and grade, and display the mark list using structure pointers.

10. File Handling

10.1 Integer Functions

1. Read and write numbers using getw() and putw().
2. Separate even and odd numbers into two files.
3. Separate prime and non-prime numbers into two files.
4. Separate palindrome and non-palindrome numbers into two files.

10.2 Character Functions

1. Read and write characters using getc() and putc().
2. Separate vowels and consonants into two files.

10.3 Mixed Data Types

1. Read and write student details using fprintf() and fscanf().